Anesthesia for Cardiovascular Diseases MCQ – Chapter 18

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 18)

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. What level of effort tolerance (in METS) is generally associated with lesser perioperative complications in patients with ischemic heart disease? **A.** > 2 METS **B.** > 4 METS **C.** > 6 METS **D.** > 8 METS	**Ans: B**
Q2. How many parameters are included in the Lee revised cardiac risk index? **A.** 4 **B.** 5 **C.** 6 **D.** 8	**Ans: C**
Q3. Regarding beta-blockers in IHD patients presenting for surgery, which statement is correct? **A.** Beta-blockers should be stopped 5 days before surgery **B.** Beta-blockers must not be stopped at any cost if the patient is already on them **C.** Beta-blockers should be freshly started on the day before surgery **D.** Beta-blockers should be replaced with calcium channel blockers preoperatively	**Ans: B**
Q4. How many days before surgery should clopidogrel be stopped in an IHD patient? **A.** 3 days **B.** 5 days **C.** 7 days **D.** 10 days	**Ans: B**
Q5. For how long should elective surgery be deferred after placement of a conventional drug eluting stent? **A.** 6 weeks **B.** 3 months **C.** 6 months **D.** 1 year	**Ans: D**
Q6. Which combination of ECG leads can detect 94–95% of intraoperative ischemic events? **A.** Lead I and aVL **B.** Lead II and V5 **C.** Lead III and V1 **D.** Lead aVR and V6	**Ans: B**

Q7. Which is the earliest and most sensitive monitor to detect intraoperative myocardial infarction? **A.** ECG with ST analysis **B.** Pulmonary artery catheter **C.** Transesophageal echocardiography (TEE) **D.** Central venous pressure monitoring	**Ans: C**

Q8. A patient with IHD and normal LV function is undergoing surgery. Which of the following should be AVOIDED during maintenance of anesthesia? **A.** Isoflurane **B.** Sevoflurane **C.** Nitrous oxide **D.** Opioids	**Ans: C**

Q9. Mitral stenosis is considered severe when the transvalvular pressure gradient exceeds: **A.** 5 mm Hg **B.** 10 mm Hg **C.** 15 mm Hg **D.** 20 mm Hg	**Ans: B**

Q10. Why should tachycardia be avoided in patients with mitral stenosis? **A.** It increases the afterload on the left ventricle **B.** It causes coronary vasoconstriction **C.** It decreases diastole, reducing left ventricular filling and hence cardiac output **D.** It increases the degree of mitral regurgitation	**Ans: C**

Q11. Approximately what fraction of mitral stenosis patients have atrial fibrillation? **A.** One-tenth **B.** One-fourth **C.** One-third **D.** One-half	**Ans: C**

Q12. Mitral regurgitation is considered severe when the regurgitant fraction exceeds: **A.** 0.3 **B.** 0.4 **C.** 0.5 **D.** 0.6	**Ans: D**

Q13. In mitral regurgitation, which hemodynamic change should be avoided? **A.** Mild tachycardia **B.** Decrease in afterload **C.** Bradycardia **D.** Mild hypotension	**Ans: C**

Q14. Which antihypertensives should be stopped on the day of surgery in hypertensive patients? **A.** Beta-blockers and calcium channel blockers **B.** ACE inhibitors and angiotensin II receptor antagonists **C.** Diuretics and alpha blockers **D.** All antihypertensives should be continued	**Ans: B**

Q15. In mitral valve prolapse (MVP), which assessment is MOST important preoperatively? **A.** Degree of left atrial enlargement **B.** Whether MVP is associated with mitral regurgitation or not **C.** Presence of atrial fibrillation **D.** Left ventricular ejection fraction	**Ans: B**

Q16. Which induction agent is the agent of choice for shock patients? **A.** Propofol **B.** Thiopentone **C.** Etomidate **D.** Ketamine	**Ans: D**

Q17. Among valvular lesions, which has the most devastating complications including sudden death? **A.** Mitral stenosis **B.** Mitral regurgitation **C.** Aortic stenosis **D.** Aortic regurgitation	**Ans: C**

Q18. Aortic stenosis is considered severe when the aortic valve orifice is less than: **A.** 1.5 cm^2 **B.** 1.0 cm^2 **C.** 0.8 cm^2 **D.** 0.5 cm^2	**Ans: C**

Q19. Which vasopressor is the agent of choice for patients with aortic stenosis, and why? **A.** Ephedrine, because it increases cardiac output **B.** Noradrenaline, because it has strong alpha effect **C.** Phenylephrine, because it does not cause tachycardia **D.** Dopamine, because it improves renal blood flow	**Ans: C**

Q20. In aortic regurgitation, which hemodynamic changes should be avoided? **A.** Tachycardia and hypotension **B.** Bradycardia and hypertension **C.** Tachycardia and hypertension **D.** Bradycardia and hypotension	**Ans: B**

Q21. Which type of prosthetic heart valve is more prone to thromboembolism but has a longer life?

A. Bioprosthetic valve (10–15 years)
B. Mechanical valve (20–30 years)
C. Homograft valve
D. Autograft valve

Ans: B

Q22. How many days before surgery should warfarin be stopped in patients with prosthetic heart valves?

A. 2 days
B. 3 days
C. 5 days
D. 7 days

Ans: C

Q23. According to new guidelines, antibiotic prophylaxis for patients with prosthetic heart valves is recommended for which procedures?

A. All surgical procedures
B. All procedures involving body cavities
C. Dental procedures involving gingiva/oral mucosa/periapical region and invasive procedures involving incision on infective tissue
D. Only cardiac surgeries

Ans: C

Q24. What is the basic aim of anesthetic management in left to right shunts (VSD, ASD, PDA)?

A. Maintain high SVR and low PVR
B. Maintain low SVR and high PVR
C. Maintain high SVR and high PVR
D. Maintain low SVR and low PVR

Ans: B

Q25. For which congenital defect is antibiotic prophylaxis for infective endocarditis recommended ONLY if there is a concomitant valvular abnormality?

A. VSD
B. ASD
C. PDA
D. Tetralogy of Fallot

Ans: B

Q26. In left to right shunts, what effect does increased pulmonary blood flow have on IV induction agents?

A. Significantly delays onset
B. Causes dilution but has no clinical implication
C. Significantly hastens onset
D. Causes unpredictable effects requiring dose reduction

Ans: B

Q27. A patient in shock must be resuscitated to at least what minimum blood pressure before considering anesthesia?

- **A.** MAP > 50 mm Hg or SBP > 80 mm Hg
- **B.** MAP > 65 mm Hg or SBP > 90 mm Hg
- **C.** MAP > 75 mm Hg or SBP > 100 mm Hg
- **D.** MAP > 80 mm Hg or SBP > 110 mm Hg

Ans: B

Q28. Which is the induction agent of choice for right to left shunts (cyanotic heart diseases)?

- **A.** Propofol
- **B.** Thiopentone
- **C.** Etomidate
- **D.** Ketamine

Ans: D

Q29. What happens to the onset of IV induction agents in right to left shunts?

- **A.** Onset is delayed
- **B.** Onset is rapid as pulmonary circulation is bypassed
- **C.** Onset is unchanged
- **D.** Onset is slower due to polycythemia

Ans: B

Q30. Which muscle relaxant is the agent of choice in right to left shunts, and why?

- **A.** Vecuronium, because it is cardiac stable
- **B.** Atracurium, because it does not affect SVR
- **C.** Pancuronium, because it causes hypertension
- **D.** Succinylcholine, because of its rapid onset

Ans: C

Q31. In right to left shunts, the FiO■ used is 50% instead of 33% because:

- **A.** Higher oxygen improves cerebral perfusion
- **B.** High FiO■ decreases PVR, which is beneficial
- **C.** It compensates for polycythemia
- **D.** N■O at higher concentrations is toxic

Ans: B

Q32. In coarctation of aorta, what is the minimum mean arterial pressure that should be maintained in the lower limb?

- **A.** > 30 mm Hg
- **B.** > 40 mm Hg
- **C.** > 50 mm Hg
- **D.** > 60 mm Hg

Ans: B

Q33. Regarding heart failure patients, which statement is correct?

- **A.** Elective surgery can proceed if LVEF > 30%
- **B.** Elective surgery is contraindicated in heart failure patients
- **C.** Only regional anesthesia is suitable
- **D.** Beta-blockers must be stopped before surgery

Ans: B

Question	Answer
Q34. What is the most common cause of right heart failure? **A.** Pulmonary embolism **B.** Chronic lung disease **C.** Left heart failure **D.** Primary pulmonary hypertension	**Ans: C**
Q35. Which drugs remain the primary treatment for chronic heart failure? **A.** Beta-blockers and calcium channel blockers **B.** Digitalis and diuretics **C.** ACE inhibitors, ARBs and aldosterone antagonists (RAAS inhibitors) **D.** Nitrates and hydralazine	**Ans: C**
Q36. Which preoperative electrolyte is particularly important to check in heart failure patients on diuretics and digitalis? **A.** Sodium **B.** Calcium **C.** Potassium **D.** Magnesium	**Ans: C**
Q37. Why is positive pressure ventilation beneficial in heart failure patients? **A.** It increases myocardial contractility **B.** It decreases venous return (preload) **C.** It increases systemic vascular resistance **D.** It improves coronary blood flow	**Ans: B**
Q38. What is the most common genetic cardiovascular disease? **A.** Dilated cardiomyopathy **B.** Hypertrophic cardiomyopathy **C.** Restrictive cardiomyopathy **D.** Arrhythmogenic right ventricular cardiomyopathy	**Ans: B**
Q39. What is the most important feature of hypertrophic cardiomyopathy (HCM)? **A.** Diastolic dysfunction **B.** Dynamic left ventricular outflow tract (LVOT) obstruction **C.** Mitral regurgitation **D.** Systolic anterior motion of the mitral valve	**Ans: B**
Q40. Pulmonary edema in HCM patients CANNOT be treated by diuretics, nitrates, and digitalis. What is the recommended treatment? **A.** Dobutamine and milrinone **B.** Phenylephrine (vasopressor) and esmolol (β-blocker) **C.** Adrenaline and atropine **D.** Furosemide and morphine	**Ans: B**

Q41. Why is pancuronium contraindicated in HCM? **A.** It causes bradycardia **B.** It can cause sympathetic stimulation which worsens LVOT obstruction **C.** It causes histamine release **D.** It has a prolonged duration of action	**Ans: B**

Q42. Which is the most common type of cardiomyopathy? **A.** Hypertrophic cardiomyopathy **B.** Dilated cardiomyopathy **C.** Restrictive cardiomyopathy **D.** Obliterative cardiomyopathy	**Ans: B**

Q43. In patients with pericardial diseases (constrictive pericarditis/cardiac tamponade), which induction agent is preferred? **A.** Propofol **B.** Thiopentone **C.** Etomidate **D.** Ketamine	**Ans: D**

Q44. In pericardial diseases, which of the following statements is correct regarding anesthetic technique? **A.** Central neuraxial blocks are the technique of choice **B.** Pericardiocentesis should be done under GA before the procedure **C.** CNB (spinal/epidural) are contraindicated and pericardiocentesis should be done under local anesthesia before GA **D.** Only monitored anesthesia care is appropriate	**Ans: C**

Q45. Which muscle relaxant is preferred for patients with pericardial diseases, and why? **A.** Vecuronium, because it is cardiac stable **B.** Pancuronium, because it stimulates sympathetic system and increases cardiac output **C.** Atracurium, because it causes histamine release **D.** Rocuronium, because of rapid onset	**Ans: B**

Q46. What is the most common indication for cardiac implanted electronic devices (CIED)? **A.** Atrial fibrillation **B.** Ventricular tachycardia **C.** Sick sinus syndrome **D.** Complete heart block	**Ans: C**

Q47. What is the most common source of electromagnetic interference (EMI) that can cause CIED malfunction during surgery? **A.** MRI **B.** Radiofrequency devices **C.** Cautery (electrocautery) **D.** Ultrasound devices	**Ans: C**

Q48. Before surgery, CIED should be programmed to asynchronous mode. In this mode, the pacemaker delivers a fixed rate of approximately: **A.** 50–55 beats/minute **B.** 60–65 beats/minute **C.** 70–72 beats/minute **D.** 80–85 beats/minute	**Ans: C**
Q49. Hypertension is defined as blood pressure exceeding which value on at least 2 occasions? **A.** > 120/80 mm Hg **B.** > 130/85 mm Hg **C.** > 140/90 mm Hg **D.** > 150/95 mm Hg	**Ans: C**
Q50. What is the general consensus for the diastolic BP threshold to delay elective surgery in hypertensive patients? **A.** > 90 mm Hg **B.** > 100 mm Hg **C.** > 110 mm Hg **D.** > 120 mm Hg	**Ans: C**

Answer Key & Explanations

Q1. Answer: B — > 4 METS

As per the text, effort tolerance > 4 METS is generally associated with lesser complications in IHD patients undergoing surgery.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

Q2. Answer: C — 6

The Lee revised cardiac risk index includes 6 parameters: high risk surgery, history of IHD, cerebrovascular disease, congestive heart failure, insulin dependent diabetes mellitus, and preoperative serum creatinine > 2 mg/dL.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

Q3. Answer: B — Beta-blockers must not be stopped at any cost if the patient is already on them

Beta-blockers must not be stopped at any cost if the patient is on them, as stopping can significantly increase morbidity and mortality. They should not be started afresh unless they can be started at least 1 week before surgery.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

Q4. Answer: B — 5 days

Clopidogrel should be stopped 5 days before surgery. However, central neuraxial block can be given only after 7 days of stopping clopidogrel.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

Q5. Answer: D — 1 year

Elective surgery must be deferred for 6 months with new generation drug eluting stents and 1 year with conventional drug eluting stents.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

Q6. Answer: B — Lead II and V5

Lead V5 is best for detecting left ventricle infarction and lead II for arrhythmia (and inferior wall MI). A combination of lead II and V5 can detect 94–95% of intraoperative ischemic events.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 183

Q7. Answer: C — Transesophageal echocardiography (TEE)

TEE is the earliest and most sensitive monitor to detect intraoperative infarction, capable of detecting more than 99% of ischemic events.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 184

Q8. Answer: C — Nitrous oxide

Nitrous oxide should be avoided in IHD patients for two reasons: it may cause mild sympathetic stimulation and it is a pulmonary vasoconstrictor. Inhalational agents can be safely used if LV function is normal.

Topic: Ischemic Heart Disease | Ref: Ch 18, Section: Ischemic Heart Disease, Page 184

Q9. Answer: B — 10 mm Hg

Mitral stenosis is considered severe if the transvalvular pressure gradient is > 10 mm Hg (normal < 5 mm Hg).

Topic: Valvular Diseases – Mitral Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 185

Q10. Answer: C — It decreases diastole, reducing left ventricular filling and hence cardiac output

In mitral stenosis, tachycardia decreases diastole which reduces left ventricular filling and hence stroke volume and cardiac output.

Topic: Valvular Diseases – Mitral Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 185

Q11. Answer: C — One-third

Atrial fibrillation is seen in one-third of the patients with mitral stenosis, and a significant number will be on oral anticoagulants contraindicating the use of CNB.

Topic: Valvular Diseases – Mitral Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 185

Q12. Answer: D — 0.6

Mitral regurgitation is said to be severe if the regurgitant fraction is > 0.6.

Topic: Valvular Diseases – Mitral Regurgitation | Ref: Ch 18, Section: Valvular Diseases, Page 185

Q13. Answer: C — Bradycardia

In MR, bradycardia should be avoided because it increases diastole which increases the regurgitation. Hypertension should also be avoided as it increases afterload.

Topic: Valvular Diseases – Mitral Regurgitation | Ref: Ch 18, Section: Valvular Diseases, Page 185

Q14. Answer: B — ACE inhibitors and angiotensin II receptor antagonists

All antihypertensives should be continued except ACE inhibitors and angiotensin II receptor antagonists, which are stopped on the day of surgery as they can cause profound intraoperative hypotension.

Topic: Hypertension | Ref: Ch 18, Section: Hypertension, Page 190

Q15. Answer: B — Whether MVP is associated with mitral regurgitation or not

The hemodynamic management principles of MVP without MR and with MR are different, so it is very important to assess preoperatively whether MVP is associated with MR or not.

Topic: Valvular Diseases – Mitral Valve Prolapse | Ref: Ch 18, Section: Valvular Diseases, Page 185-186

Q16. Answer: D — Ketamine

Ketamine is the induction agent of choice for shock patients due to its sympathomimetic effects. However, in chronic/severe shock with sympathetic depletion, its direct myocardial depressant effect may predominate.

Topic: Hypotension/Shock | Ref: Ch 18, Section: Hypotension/Shock, Page 190-191

Q17. Answer: C — Aortic stenosis

Among the valvular lesions, aortic stenosis has the most devastating complications like infarction, arrhythmia or even sudden death.

Topic: Valvular Diseases – Aortic Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 186

Q18. Answer: C — 0.8 cm^2

Severe AS is defined as transvalvular gradient > 50 mm Hg or aortic valve orifice less than 0.8 cm^2.

Topic: Valvular Diseases – Aortic Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 186

Q19. Answer: C — Phenylephrine, because it does not cause tachycardia

As phenylephrine does not cause tachycardia, it is the vasopressor of choice for patients with aortic stenosis.

Topic: Valvular Diseases – Aortic Stenosis | Ref: Ch 18, Section: Valvular Diseases, Page 186

Q20. Answer: B — Bradycardia and hypertension

In AR, bradycardia by increasing diastole and hypertension by increasing afterload can increase the regurgitant fraction, so both should be avoided.

Topic: Valvular Diseases – Aortic Regurgitation | Ref: Ch 18, Section: Valvular Diseases, Page 186

Q21. Answer: B — Mechanical valve (20–30 years)

Mechanical prosthetic heart valves are more prone for thromboembolism but have a long life of 20–30 years, while bioprosthetic valves are less prone for thromboembolism but have shorter life of 10–15 years.

Topic: Prosthetic Heart Valves | Ref: Ch 18, Section: Prosthetic Heart Valves, Page 186

Q22. Answer: C — 5 days

Warfarin should be stopped 5 days before surgery and heparin should be started, which is then stopped 12–24 hours before surgery.

Topic: Prosthetic Heart Valves | Ref: Ch 18, Section: Prosthetic Heart Valves, Page 186

Q23. Answer: C — Dental procedures involving gingiva/oral mucosa/periapical region and invasive procedures involving incision on infective tissue

The new guidelines recommend antibiotic prophylaxis only for dental procedures involving gingiva, oral mucosa and periapical region of teeth, and invasive procedures involving incision on infective tissue.

Topic: Prosthetic Heart Valves | Ref: Ch 18, Section: Prosthetic Heart Valves, Page 186

Q24. Answer: B — Maintain low SVR and high PVR

In left to right shunts, increase in SVR or decrease in PVR increases the shunt fraction. The aim is to maintain low systemic vascular resistance and high pulmonary vascular resistance.

Topic: Congenital Heart Diseases – Left to Right Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q25. Answer: B — ASD

Antibiotic prophylaxis is recommended for VSD and PDA, but for ASD only if there is a concomitant valvular abnormality (mitral valve prolapse may be associated with ASD).

Topic: Congenital Heart Diseases – Left to Right Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q26. Answer: B — Causes dilution but has no clinical implication

Although increase in pulmonary blood flow causes dilution of IV agents in left to right shunts, this has no clinical implication.

Topic: Congenital Heart Diseases – Left to Right Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q27. Answer: B — MAP > 65 mm Hg or SBP > 90 mm Hg

Patient must be resuscitated to at least mean arterial pressure > 65 mm Hg or systolic pressure > 90 mm Hg before considering anesthesia.

Topic: Hypotension/Shock | Ref: Ch 18, Section: Hypotension/Shock, Page 190-191

Q28. Answer: D — Ketamine

As ketamine increases the systemic vascular resistance, it is the agent of choice for induction in right to left shunts.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q29. Answer: B — Onset is rapid as pulmonary circulation is bypassed

In right to left shunts, as pulmonary circulation is bypassed, the induction with IV agents will be rapid.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q30. Answer: C — Pancuronium, because it causes hypertension

Pancuronium causes hypertension (increases SVR), which is desirable in right to left shunts to reduce shunt fraction, making it the muscle relaxant of choice.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q31. Answer: B — High FiO■ decreases PVR, which is beneficial

High FiO■ decreases PVR, which is beneficial in right to left shunts (aim is to decrease PVR), so oxygen concentration used is 50%.

Topic: Congenital Heart Diseases – Right to Left Shunts | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q32. Answer: B — > 40 mm Hg

In coarctation of aorta, blood pressure monitoring in lower limbs is mandatory and mean arterial pressure of the lower limb should remain at least > 40 mm Hg.

Topic: Congenital Heart Diseases – Coarctation of Aorta | Ref: Ch 18, Section: Congenital Heart Diseases, Page 187

Q33. Answer: B — Elective surgery is contraindicated in heart failure patients

The patient in heart failure carries very high risk of perioperative morbidity and mortality, therefore elective surgery is contraindicated.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

Q34. Answer: C — Left heart failure

The most common cause of right heart failure is left heart failure.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

Q35. Answer: C — ACE inhibitors, ARBs and aldosterone antagonists (RAAS inhibitors)

The RAAS system plays an important role in heart failure pathophysiology. ACE inhibitors, ARBs and aldosterone antagonists remain the primary drugs for treatment of chronic heart failure.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

Q36. Answer: C — Potassium

As heart failure patients are on diuretics and digitalis, checking preoperative electrolytes, especially potassium, is a must.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

Q37. Answer: B — It decreases venous return (preload)

Positive pressure ventilation is beneficial in heart failure by decreasing the venous return (preload), which is one of the basic principles of heart failure management.

Topic: Heart Failure | Ref: Ch 18, Section: Heart Failure, Page 188

Q38. Answer: B — Hypertrophic cardiomyopathy

Hypertrophic cardiomyopathy (HCM), also called subaortic stenosis, is the most common genetic cardiovascular disease.

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188

Q39. Answer: B — Dynamic left ventricular outflow tract (LVOT) obstruction

Dynamic left ventricular outflow obstruction (LVOT) is the most important feature of HCM.

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188

Q40. Answer: B — Phenylephrine (vasopressor) and esmolol (β-blocker)

In HCM, pulmonary edema cannot be treated by diuretics (decrease preload), nitrates (decrease afterload), or digitalis (increase contractility) as all worsen LVOT obstruction. It is treated by vasopressors (phenylephrine) and β-blockers (esmolol).

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188-189

Q41. Answer: B — It can cause sympathetic stimulation which worsens LVOT obstruction

Pancuronium is contraindicated in HCM because it can cause sympathetic stimulation, which increases myocardial contractility and worsens LVOT obstruction.

Topic: Cardiomyopathies – HCM | Ref: Ch 18, Section: Cardiomyopathies, Page 188

Q42. Answer: B — Dilated cardiomyopathy

Dilated cardiomyopathy is the most common cardiomyopathy. It may be primary (idiopathic) or secondary like alcoholic or peripartum cardiomyopathy.

Topic: Cardiomyopathies – DCM | Ref: Ch 18, Section: Cardiomyopathies, Page 189

Q43. Answer: D — Ketamine

In pericardial diseases, ketamine is preferred for induction as it stimulates the sympathetic system and increases cardiac output in these low cardiac output states.

Topic: Pericardial Diseases | Ref: Ch 18, Section: Pericardial Diseases, Page 189

Q44. Answer: C — CNB (spinal/epidural) are contraindicated and pericardiocentesis should be done under local anesthesia before GA

Patients with pericardial diseases are in low cardiac output state, so CNB are contraindicated. Pericardiocentesis should be done under local anesthesia before proceeding to GA.

Topic: Pericardial Diseases | Ref: Ch 18, Section: Pericardial Diseases, Page 189

Q45. Answer: B — Pancuronium, because it stimulates sympathetic system and increases cardiac output

Pancuronium is preferred for pericardial diseases because it stimulates the sympathetic system and increases cardiac output, which is needed in the low cardiac output state.

Topic: Pericardial Diseases | Ref: Ch 18, Section: Pericardial Diseases, Page 189

Q46. Answer: C — Sick sinus syndrome

CIED are used for treating bradyarrhythmias, with sick sinus syndrome being the most common indication.

Topic: Cardiac Implanted Electronic Devices | Ref: Ch 18, Section: CIED/Pacemakers, Page 189

Q47. Answer: C — Cautery (electrocautery)

The most common source of EMI is cautery, followed by radiofrequency devices and MRI. MRI is contraindicated for patients with CIED.

Topic: Cardiac Implanted Electronic Devices | Ref: Ch 18, Section: CIED/Pacemakers, Page 189

Q48. Answer: C — 70–72 beats/minute

In asynchronous mode, pacemakers deliver a fixed rate of 70–72 beats/minute irrespective of the patient's own rate, avoiding the risk of CIED being affected by interference.

Topic: Cardiac Implanted Electronic Devices | Ref: Ch 18, Section: CIED/Pacemakers, Page 189

Q49. Answer: C — > 140/90 mm Hg

Hypertension is defined as blood pressure > 140/90 on at least 2 occasions 1–2 weeks apart, while hypertensive crisis is defined as BP > 180/120.

Topic: Hypertension | Ref: Ch 18, Section: Hypertension, Page 190

Q50. Answer: C — > 110 mm Hg

Although there is no universal standard cut-off, the general consensus is to delay elective surgery if diastolic blood pressure is > 110 mm Hg.

Topic: Hypertension | Ref: Ch 18, Section: Hypertension, Page 190